

Ontario Cancer Hospital

Division of Tumor
Sequencing and
Diagnostics

123 Main St. West,
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fax: 416-456-7890
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Draft LABORATORY RESULTS

Patient Info:

Name:
DOB:
Sex:

Patient Information:

Physician Name:
Health Card:
MRN #:
Clinic: Peterborough Regional
Health Centre (Peterborough)

Procedure Date
2023-12-24 12:35:00

Accession Date
2023-12-25 12:46:26

Report Date
2023-12-26 21:35:43

Report Details

Specimen Tested - Amplified DNA

Sequencing Scope - Whole genome sequencing (WGS)

Referral Reason - Clinical

Table of findings:

Gene	Information	Interpretation
PDGFRB	1 c.3133G>C p.Ala1045Pro heterozygous	Variant of uncertain clinical significance
PCM1	8 c.5741T>G p.Leu1914Ter heterozygous	Variant of uncertain clinical significance

Summary: Two variants of uncertain clinical significance detected.. The details regarding the specific mutations are included below.

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Variant Interpretation:

The interpretation of these variants is as follows: Two variants of uncertain clinical significance were detected in the sample.

Variant 1 of 2

Gene	Variant	Amino	Zygoty
PDGFRB	c.3133G>C	p.Ala1045Pro	heterozygous

Variant of uncertain clinical significance

PDGFRB c.3133G>C p.Ala1045Pro begins at position 1045 in exon 1 within chromosome chr5 . This mutation has been identified in 33 families. It causes an amino acid substitution, which replaces alanine with proline . According to ClinVar, the evidence collected to date is insufficient to firmly establish the clinical significance of this variant, therefore it is classified as a variant of uncertain clinical significance . ClinVar and other genomic databases report the PDGFRB c.3133G>C variant as clinically relevant based on aggregated evidence.

The clinical implications of this variant are not yet fully understood. At present, available data is insufficient to confirm its role in disease.

The affected nucleotide lies within a region that is highly conserved across vertebrate species, which suggests functional importance and evolutionary constraint.

This variant is not currently strongly implicated in specific diseases according to ClinVar records (VCV accession: VCV003752413).

Supporting studies and case reports can be found in the scientific literature. Relevant PubMed references include: 264391241, 408149228, 392324027, 391504230, 131990704, 634818416 .

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Variant 2 of 2

Gene PCM1	Variant c.5741T>G	Amino p.Leu1914Ter	Zygoty heterozygous
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Variant of uncertain clinical significance

PCM1 c.5741T>G p.Leu1914Ter begins at position 1914 in exon 8 within chromosome chr8. This mutation has been identified in 42 families. It causes an early translation termination at position 1914. According to ClinVar, the evidence collected to date is insufficient to firmly establish the clinical significance of this variant, therefore it is classified as a variant of uncertain clinical significance. ClinVar and other genomic databases report the PCM1 c.5741T>G variant as clinically relevant based on aggregated evidence.

The clinical implications of this variant are not yet fully understood. At present, available data is insufficient to confirm its role in disease.

The affected nucleotide lies within a region that is highly conserved across vertebrate species, which suggests functional importance and evolutionary constraint.

This variant is not currently strongly implicated in specific diseases according to ClinVar records (VCV accession: VCV005700711).

Supporting studies and case reports can be found in the scientific literature. Relevant PubMed references include: 691412297, 875519342, 461044787, 470172422.

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Test Details:

Genes Analyzed: SS18, CD70, CARD11, SDHB, BCL6, KEAP1, TFPT, RBM15, NCOR2, CREB1, CD79A,

Recommendations

A precision oncology approach is advised. These mutations are known oncogenic drivers, linked to constitutive pathway activation. Targeted therapies, including hormone-correcting agents, may be considered, guided by clinical judgment. PI3K inhibitors could be explored in trials for PIK3CA-mutated cases. Germline testing is not indicated, as all mutations are consistent with somatic events. A multidisciplinary tumour board review is recommended to integrate findings into care. Additional testing may be pursued at the physician's discretion.

Methodology

Genomic DNA was analyzed using a targeted panel covering all coding exons and adjacent intronic regions. Target enrichment was performed with hybrid capture (Twist Bioscience), followed by Illumina NextSeq sequencing. Reads were aligned to GRCh37 using BWA-MEM, and variants called with GATK. Annotation was performed in VarSeq using population databases, predictive algorithms, and ClinVar. CNVs were assessed with CNVkit and confirmed by MLPA when applicable. Regions with pseudogene interference, such as PMS2, were validated using long-range PCR and Sanger sequencing. Mean read depth exceeded 300x, with a minimum threshold of 50x. Analytical sensitivity is >99% for SNVs/indels and >95% for exon-level CNVs. **Only variants classified as pathogenic, likely pathogenic, or variants of uncertain significance (VUS) are reported**, per ACMG/AMP guidelines (PMID: 25741868).

Limitations

This test was developed and validated in a certified clinical laboratory. Limitations include reduced sensitivity in pseudogene regions (e.g., PMS2, CHEK2), and inability to detect certain structural variants (e.g., MSH2 inversions), deep intronic changes, or low-level mosaicism. PMS2 exons 11–15 are not analyzed. Interpretations reflect current knowledge and may be updated as new evidence emerges.

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Report Electronically Verified and Signed
by:

Mock specimen